

Complex Systems, Costly Change: How Functional Feedback Shapes Policy Punctuations

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Policy change often follows patterns of stability interrupted by significant shifts, as described by punctuated equilibrium theory (PET). This paper explores how dynamic feedback mechanisms, specifically functional costs, shape these policy punctuations within complex legislative environments. A theoretical model is developed, integrating functional costs — such as drafting, harmonization, and compliance expenses — into a dynamic optimization process guiding policy-making. Simulations show that functional costs act as dynamic feedback evolving over time, explaining inter-subsystem variance in policy punctuations. This argument is tested using quantile regression and network analysis on EU policy outputs from 1980 to 2015. The findings reveal that higher functional costs within a policy subsystem significantly reduce the likelihood of extreme policy changes. Leveraging variance in the relevance of functional costs across legal instruments provides further support for the proposed mechanism. This study highlights the importance of subsystem complexity and functional costs in policy stability and change, offering new insights into the mechanisms underlying policy punctuations and the evolution of complex policy systems.

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1 Introduction

Policy change is a complex and dynamic phenomenon. Understanding what triggers or inhibits change is crucial for policy-makers, researchers, and practitioners who seek to address societal problems and improve policy outcomes. Consequently, various theoretical frameworks and models attempt to explain how different factors can interact and influence policy change over time. Some of these frameworks focus on specific aspects, such as institutions ([North 1990a](#); [Pierson 2000](#); [Thelen 2004](#)), interests ([Sabatier 1988](#); [Kingdon 1995](#)), ideas ([Hall 1993](#); [Béland 2009](#); [Schmidt 2010](#)), networks ([Rhodes 1997](#); [Klijn, Steijn, and Edelenbos 2010](#); [Cairney 2016](#)), or events ([Birkland 2006](#)). Others try integrating multiple factors into a comprehensive approach, such as punctuated equilibrium theory ([Baumgartner and Jones 1993](#)).

At its core, PET describes systems that produce a pattern of change that inhibits long periods of stasis occasionally interrupted by brief periods of meaningful and substantial change. Baumgartner and Jones ([1993](#)) adopted the theory stemming from paleobiology ([Gould and Eldredge 1977](#)) to describe the patterns observable in the change of public agendas. While initially focused on explaining stability and change, PET has been extended to a general information processing theory ([Jones and Baumgartner 2004](#)). The theory suggests that policy systems are confronted with policy inputs that consist of nearly infinite components. These inputs can be assumed to sum up to normally distributed input demands (see also [Padgett 1980](#)). Although this assumption has been challenged ([Desmarais 2019](#)), staying with it, a perfect responsive system should produce normally distributed outputs from normally distributed inputs.

Friction and positive or negative feedback are proposed mechanisms for why this is not the case ([Jones and Baumgartner 2004](#)). Yet, while the effects of friction have been extensively studied (e.g., [Baumgartner et al. 2009](#)), more general feedback mechanisms have been underexplored. In theory, positive feedback mechanisms amplify signals given by policy inputs, leading to punctuations, while negative feedback mechanisms inhibit the response to inputs, causing incremental change patterns ([Jones and Baumgartner 2004](#)). Although a central mechanism within the explanation of punctuated patterns, most empirical studies do not directly engage with these feedback mechanisms attributed to creating punctuations ([Kuhlmann and van der Heijden 2018](#)).

This paper argues that the focus on decision costs, primarily tied to budgetary changes, has overshadowed other critical factors like the costs of drafting, harmonization, and compliance,

particularly in the context of legislative outputs in complex settings. This has constrained theoretical contributions and empirical investigations. Furthermore, while PET aligns with complex systems theory (Jones, Sulkin, and Larsen 2003; Jones and Baumgartner 2012), few studies have probed how feedback mechanisms from complex systems influence policy change (Stauffer et al. 2022), leaving a gap in understanding the connection between policy inputs and outputs (Desmarais 2019) and the role of inter-subsystem dynamics (Jochim and May 2010).

In response to these gaps, this paper introduces a theoretical model and provides empirical evidence to demonstrate how dynamics across policy subsystems generate feedback mechanisms that accelerate or inhibit policy punctuations. By introducing the concept of dynamic feedback, derived from complex systems theory, it offers an additional explanation for punctuations in policy outputs.

Dynamic feedback recognizes that policy-makers face temporal bound challenges when producing policy outputs that exhibit different dynamics over time. It highlights that functional constraints alone can lead to punctuated output without external influences such as crisis events. To test the theoretical mechanism developed, I analyze the change distributions of 30 years of policy outputs from the European Union between 1975 and 2015. The mechanism is tested using quantile regression, with measures retrieved from a network analysis of the legal landscape to capture functional costs policy-makers face. The results indicate that higher functional costs in a given year decrease the likelihood of punctuation occurring in subsequent periods.

The paper is structured as follows: Section 2 discusses the role of costs in punctuated equilibrium theory (PET), focusing on functional costs. Section 3 introduces the theoretical model and demonstrates how it can simulate policy-making under information constraints. Section 5 outlines the empirical strategy used to identify the effect of functional costs on policy punctuations in EU policy-making. Section 6 presents the results, and Section 7 concludes the study by summarizing the findings and discussing their implications.

2 Costs in policy making

When examining patterns of change, the literature on policy punctuations commonly emphasizes the role of political institutions and the extent to which these institutions encounter

costs that may create obstacles in the policy-making process (Robinson et al. 2007; Baumgartner et al. 2009; Jones et al. 2009; Breunig 2011). Jones, Sulkin, and Larsen (2003) categorize the costs faced by decision-making systems into four types: decision, transaction, information, and cognitive costs.

Linking to Buchanan and Tullock (1965), they define decision costs as costs that occur for reaching an agreement across political actors. While they separate transaction and informational costs, I argue in line with North (1990b) that they are highly intertwined. They state: “The costliness of information is the key to the costs of transacting, which consist of the costs of measuring the valuable attributes of what is being exchanged and the costs of protecting rights and policing and enforcing agreements.” (p. 27). Regarding inhibiting change, one can think of these costs as costs that arise due to the functional nature of policy systems. Hence, in this paper, I will refer to them as *functional costs*. Lastly, they define cognitive costs as stemming from the limited processing capacity inherent in any social institution that is comprised of human individuals. This reflects the bounded rationality approach adopted to PET (see also Simon 1957; Padgett 1980).

In terms of relevancy for them, “[...] costs of bringing relevant parties to agreement in a system of separated powers (decision costs) generally outweigh the costs of holding hearings, enacting statutes, or changing budgetary allocations once agreement has been reached (transaction costs).” (p. 154). This string of argumentation has two caveats.

First, Jones, Sulkin, and Larsen (2003) rest their argument on the institutional level. Hence, for them, the decision-making procedure of the institution and, most importantly, the *differences* in decision-making procedures across institutions are the primary source of cost differences that might explain differences in punctuated patterns (see also Workman, Jones, and Jochim 2009). Yet, this distinction has no explanatory power when the focus is on dissecting *within* variance of institutional outputs and explaining individual punctuations, especially regarding legislative outputs. Here, we can assume that cognitive and even decision costs stemming from the institutional setup are relatively stable (Mortensen 2009). Although Spiller and Tiller (1997) show that decision costs under certain conditions can be manipulated as well.

The latter part of the statement highlights a secondary limitation: the assumption that functional costs only materialize after an agreement has been reached does not fully account for their presence at multiple stages throughout the policy-making process (Krutilla 2011). Scholars such as Schuck (1992) argue that functional costs are a persistent feature at various stages of policy development. According to them, a law with higher associated costs

“[...] tends to be more costly and cumbersome to administer, more difficult for lawmakers to formulate and agree upon, and more difficult to reform once established” (p. 18). Consequently, Hurka and Haag (2020) demonstrate that the length of legislative processes can be significantly extended with rising costs, while Haag, Hurka, and Kaplaner (2024) find that the success of policy implementation is heavily dependent on them.

Therefore, functional costs can be understood as reflecting the resources required to navigate the complexities of the existing legal and regulatory framework when developing new policies. This includes costs associated with drafting legislation, ensuring consistency with existing laws, harmonizing regulations across different sectors or jurisdictions, and even implementing and enforcing new policies.

Functional costs arise from the functional nature of policy systems — they are inherent in the processes and structures required to develop and implement policies effectively. These costs are dynamic and can fluctuate based on the complexity of the policy environment, the interdependencies among existing regulations, and the administrative capacity of the institutions involved (see also Simon 1962; Pierson 2000).

This perspective challenges the traditional focus of early policy punctuation research on decision costs, which is primarily derived from a focus on budgetary changes and their relatively straightforward transmission. However, when it comes to legislative decisions, the reality is considerably more complex. These decisions often represent just the starting point or even a repeated process, with subsequent requirements to ensure compliance with the existing legal framework — a costly process making it far from trivial.

To illustrate, consider the process of drafting new financial regulations aimed at preventing systemic risk in the banking sector. Policy-makers must navigate a web of existing national and international economic rules, balance central bank policies, and align with the Basel III framework in the case of the EU. Functional costs in this case include:

- Extensive legal research to ensure compatibility with current financial laws.
- The involvement of multiple regulatory bodies to harmonize banking standards across borders.
- Consultations with industry experts to avoid unintended disruptions in capital flows.

These functional costs differ from political costs (e.g., gaining support from various interest groups) and cognitive costs (e.g., understanding the intricacies of complex financial instruments). They represent the operational effort required to develop policies that can fit within the intricate and highly interconnected regulatory environment governing global finance.

Figure 1 illustrates these complexities. It shows the legal landscape of economic policies in the European Union in 2015, with each node representing a legislative output and each edge representing references to other legislative outputs. This shows that the legal landscape is often highly complex and interdependent. Consequently, transmitting decisions into this legal framework and or designing policies that fit into it in the first place is very costly. It has the potential to affect the policy-making process significantly. Furthermore, every policy added to the system has the potential to influence the number of functional costs for future policy-making endeavors (see Pierson 2004), making the system highly dynamic. This highlights the need for a more thorough understanding of functional costs, recognizing them as integral and ongoing throughout the policy cycle.

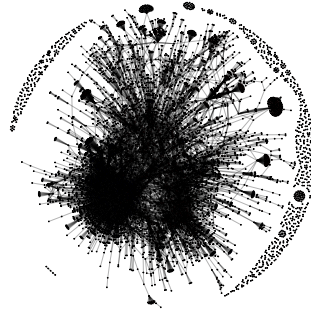


Figure 1: Legislative network of economic policies in the EU in the year 2015. Data from Fjølseth (2019)

3 Models of policy change

The strong emphasis on decision costs has also permeated models that describe how systems produce punctuated outputs. Specifically, the model by Jones and Baumgartner (2004) considers costs as a form of friction. This friction functions as a stick-slip mechanism, influencing the system's dynamics.

Their model takes an input, denoted as S_t , from a standard normal distribution and produces a response, denoted as R_t .

The transformation of inputs into responses hinges on three parameters¹:

¹The original paper uses the denominator C for the friction parameter. To avoid confusion with the functional cost parameter further down, I will refer to it as F.

F = friction

λ = efficiency

β = amplification

The response is given by R_t :

$$R_t = \begin{cases} \beta S_t & \text{if } F < (S_t + \sum S_{0 < k}) \\ \lambda \beta S_t & \text{else} \end{cases} \quad (1)$$

The dynamic aspect of the model pertains only to the $S_t + \sum S_{0 < k}$ mechanism. Essentially, in their model, inputs are only partially fulfilled if they do not cross the threshold of the stable friction parameter F since they are multiplied by the efficiency parameter λ that is bound between 0 and 1 ($0 < \lambda < 1$).

$\sum S_{0 < k}$ represents the sum of unfulfilled input signals from the prior periods and is added to the current input signal S_t .

Consequently, their model suggests that inputs are only partially realized unless they surpass the friction threshold F , being then multiplied by the efficiency parameter λ (where $0 < \lambda < 1$). When the sum of inputs exceeds the friction threshold, inputs are either completely fulfilled or amplified by the parameter β , leading to punctuated outputs.

This interpretation of friction aligns closely with the emphasis on decision costs ([Workman, Jones, and Jochim 2009](#)). Indeed, if decision costs represent the predominant costs, one might envision them functioning in a stick-slip manner. Once a decision is reached and the friction is fully overcome, punctuations are likely to occur, signaling abrupt shifts in output or policy.

3.1 Limitations of previous models

However, the model exhibits significant shortcomings in explaining how functional costs impact policy punctuations. First, it is improbable that these costs behave like friction. The major difference is that they can not be overcome suddenly since they are functional in nature. For example, policy-makers can not just ignore other provisions in the legal landscape when drafting new ones. Instead, I propose that they act as negative feedback

on policy demands, moderating or restraining changes rather than facilitating and/or amplifying them (Pierson 2000).

Second, the model focuses primarily on the institutional level, limiting its capacity to account for how cross-sectoral dynamics within institutions could influence policy-making patterns (Jochim and May 2010). This limitation makes applying the model to more diverse policy environments hard, where policy-makers have certain flexibility to choose when and where to react to policy demands. Lastly, while the model introduces a dynamic input mechanism, it statically treats costs. This assumption needs to be revised, particularly concerning dynamic functional costs, which can fluctuate based on numerous factors. A more realistic assumption would involve a dynamic integration of costs, reflecting their varying impacts over time.

3.2 A dynamic stochastic model of policy outcomes under information constraints

Therefore, I propose to focus on a dynamic stochastic model that accounts for functional costs and their cross-sectoral variance and models the decision-making as a dynamic optimization process where policy-makers try to optimize the policy outputs across multiple sectors by incorporating considerations based on demand and costs (see also Bendor, Moe, and Shotts 2001). This model aligns with complex systems theory, where policy-making systems are viewed as adaptive. Functional costs serve as dynamic feedback mechanisms, creating emergent patterns in policy outputs that reflect complex systems' non-linear and interdependent nature. I show in the following that this model can explain punctuated outputs without friction parameters.

The model contains the following main parameters:

$$C = Cost$$

$$D = Demand$$

$$U = Utility$$

Each of these parameters varies across time (t) and field (i). Additionally, there is a set of parameters which I will call modifiers:

$$x = \text{scale}$$

$$\phi_{t,i} = \text{coverage}$$

Similar to prior models, demand for a given field at a given time point ($D_{t,i}$) is defined by demand drawn from a normal distribution for each period plus the unfulfilled demands from the prior period ($D_{t-1,i}$). The unfulfilled demand represents the demand from the previous period minus the sum of fulfilled policy outputs ($\sum P_{t-1,j}$) which gets multiplied with a coverage parameter ϕ . ϕ defines the extent to which the demand was met in previous periods.

$$D_{t,i} = N(0, x) + (D_{t-1,i} - \sum P_{t-1,j} * \phi_{t-1,i}) \quad (2)$$

The initial functional cost ($C_{0,i}$) of a field is drawn from a normal function with the standard deviation of $\frac{1}{x}$. The scale parameter x captures the relationship in scale between costs and demand. I consider functional costs to be strictly positive; thus, the model takes the absolute value of the draw. Therefore, contrary to institutional setups or the like, which can act in both directions, enabling or hindering change in a given field, functional costs are initially strictly positive in the sense that costs always exist and inhibit the potential to create policies.

$$C_{0,i} = |N(0, \frac{1}{x})| \quad (3)$$

To calculate the costs for a specific time and domain ($C_{t,i}$), the model incorporates both initial costs and the costs associated with adding policies to the policy stock. When policy-makers opt to expand the policy portfolio, they inherently raise the cost for the domain, as they must accommodate the impact of each realized policy when considering future policy decisions. Consequently, each new policy can escalate forthcoming costs (Schuck 1992). Although theoretically policy-makers could opt to dismantle current policies, I operate under the assumption, supported by Adam et al. (2019), that policy-makers are strongly incentivized to accumulate new policies rather than dismantle existing ones.

In the model, the added cost is variable and results from multiplying a draw from the same normal function as before with the number of policy items realized in the previous period. Individual policies are given by the binary variable $P_{t,i,j}$. Essentially, $P_{t,i,j}$ denotes if a

particular policy item was realized ($= 1$) or was not realized ($= 0$). Therefore, $\sum P_{t-1,i,j}$ denotes the number of realized policies in a given field for the previous period.

$$C_{t,i} = C_{t-1,i} + |N(0, \frac{1}{x})| * \sum P_{t-1,i,j} \quad (4)$$

Lastly, the utility of realizing a policy $U_{t,i,j}$ in a given field for a given time is calculated by taking the difference between the demand and costs. This captures the core thought that policies that are cheap to implement but have strong demand have more utility for the policy-makers than costly policies without much demand attached. Additionally, $\phi_{t,i,j}$ represents the coverage parameter that denotes how much of the demand was fulfilled by the implemented policies.

$$U_{t,i,j} = D_{t,i} - C_{t,i} - \sum P_{t,i,j} * \phi_{t,i,j} \quad (5)$$

How policy-makers optimize given these conditions is essential to the question of whether this setup creates punctuated output distributions. For this, I will consider a limited capacity model. Under this assumption, policy-makers have a limited number of policies they can output. The total number of policies fulfilled across all fields per period is bounded between 0 and the system's capacity².

$$0 \leq \sum P_{t,i,j} < capacity \quad (6)$$

Given this limitation, policy-makers try to find the optimal solution (ω_t) by considering different utility sets. If policy-makers would have complete information about the expected values, the solution to the problem would be straightforward:

$$\omega_t = \max \sum U_{t,i,j} \quad (7)$$

The primary challenge for policy-makers within this framework stems from the incomplete information available to them and, consequently, their bounded rationality (Simon 1957; Padgett 1980). Specifically, while policy-makers might have a clear understanding of the

²In the following, I define capacity as the number of policies a system can produce within a given period. Although I do not explicitly model how more costly policies impact this capacity, the assumption is implicitly incorporated through their lower utility, as costlier policies are less likely to be selected. For further details, see Section 4.

demand for a new policy ($D_{t,i}$) and its associated costs ($C_{t,i}$) since they are aware of the policy environment in place at the time, they lack comprehensive details regarding the utility ($U_{t,i,j}$) that each policy would yield. This is because the utility's calculation hinges on the actual outcomes of policy implementation, represented by the product of policy output ($P_{t,i,j}$) and its coverage parameter ($\phi_{t,i,j}$), to determine if the policy demand has been adequately met.

In essence, while policy-makers are informed about the costs and demand for new policies, they remain uncertain about how much a specific policy will fulfill the existing demand. As a result, they are constrained to optimizing the expected total utility ($\sum \hat{U}_{t,i,j}$), reflecting their best attempt to address policy demands under the limitation of partial information.

$$\hat{U}_{t,i,j} = P_{t,i,j} * (D_{t,i} - C_{t,i}) + \epsilon \quad (8)$$

Unlike simpler models that rely on static parameters (like friction), this approach emphasizes the dynamic optimization process of policy-makers. They must continuously adapt their strategies based on changing demands and the accumulated costs of past decisions, reflecting the inherently adaptive and feedback-driven nature of complex systems. Moreover, the integration of functional costs as dynamic variables highlights how constraints are not merely static barriers but are actively influenced by prior outputs and demands in an ongoing manner. The model's structure, allowing for fluctuating costs based on past actions, ensures that these costs act as evolving feedback rather than fixed friction points. This dynamic interplay mirrors real-world policy environments, where interdependencies and the effects of prior decisions influence current and future policy choices, thus embodying the essence of complex systems theory.

4 Simulation

I use a simulation approach to test if punctuated patterns emerge given the model above. As highlighted by Sterman (2000), the benefit of simulations is that they allow us to model dynamic systems where analytical solutions may not be feasible due to the complexity and interdependence of variables involved. Since punctuations are inherently complex patterns arising from the interplay of feedback loops, functional costs, and varying demands across policy fields, solving the model analytically would require overly simplifying assumptions that may not capture the system's full dynamics. On the other hand, simulations provide

a flexible approach to explore the decision process under varying conditions, allowing the model to operate with all its constraints and interdependencies intact. This method enables me to investigate how policy outputs vary over time and under the constraints outlined above, revealing the precise mechanisms that produce punctuations or periods of stability without reducing the model’s complexity.

4.1 Assumptions

So, how do policymakers choose policy outputs under these settings? What expectations can we derive from the model above?

First, I must state the assumption under which I simulate the decision-making. The assumptions are as follows:

- **(A1) Coverage Knowledge Post-Implementation:** The value of $\phi_{t,i,j}$ is only known after policies have been realized (Simon 1957; Padgett 1980).
- **(A2) Continuous Policy Production:** Policy makers are inclined to constantly produce policy (Adam et al. 2019).
- **(A3) Costly and Limited Information Sampling:** $\sum U_{t,I,j}$ can only be gauged when observed; this process is costly and, hence, limited.
- **(A4) Capacity Constraints:** Capacity to output policy is limited but used fully (Baumgartner and Jones 1993).

Therefore, policy-makers try to fill up a constant amount of policy output activity across the policy fields. Furthermore, I assume that the total value $\sum U_{t,i,j}$ can only be gauged by policy-makers by sampling different sets of $U_{t,i,j}$ and is not a priori known. Since this process is costly, policy-makers can only inspect a finite set of samples before deciding on the set of ω_t that optimizes $\sum U_{t,i,j}$. This constraint limits the possibility of policy-makers focusing solely on one field, as they cannot exhaustively evaluate every possible scenario or concentrate all resources on a single domain. Instead, they must operate under bounded rationality, diversifying their attention across multiple fields to maximize overall utility within the constraints of limited information and evaluation costs.

Nevertheless, given the optimization process and the role of costs in determining utility, fields with lower costs inherently have a higher likelihood of being selected. When faced with similar demand levels across fields, policy-makers are more inclined to focus on those

with lower associated costs, as these provide higher utility and are less resource-intensive to implement. This tendency aligns with their objective to optimize outputs efficiently while minimizing expenditure, ensuring that fields with manageable costs and high utility are prioritized within the constraints of the sampling process. I integrate this decision approach by allowing policy-makers to construct possible solutions repeatedly and gauging their expected utility before deciding to realize a set of policies.

What patterns can be expected under these assumptions? Given that the number of policies is fixed, I expect high demand and, for the time period, more policies allocated to low-cost fields. In turn, this allocation has the potential to raise the field's costs in the following periods, decreasing the chance of high allocations. This dynamic should result in a punctuation pattern, with few periods of high allocation and many periods of stasis while other fields catch up in costs.

4.2 Scenario and results

The scenario contains $i = 1, 2, \dots, 20$ policy fields and spans a period of $t = 1, \dots, 30$. For the demand and cost development, I set $x = 10$. The coverage parameter is defined as $\phi_{i,t} = U(0, 1)$. Capacity is limited to an output of 100 policies per period and policy makers sample 100 different utility sets by drawing random combinations of policies from different fields before deciding on the highest utility one.

Based on the assumptions, I define the simulation function as presented in [Figure 2](#).

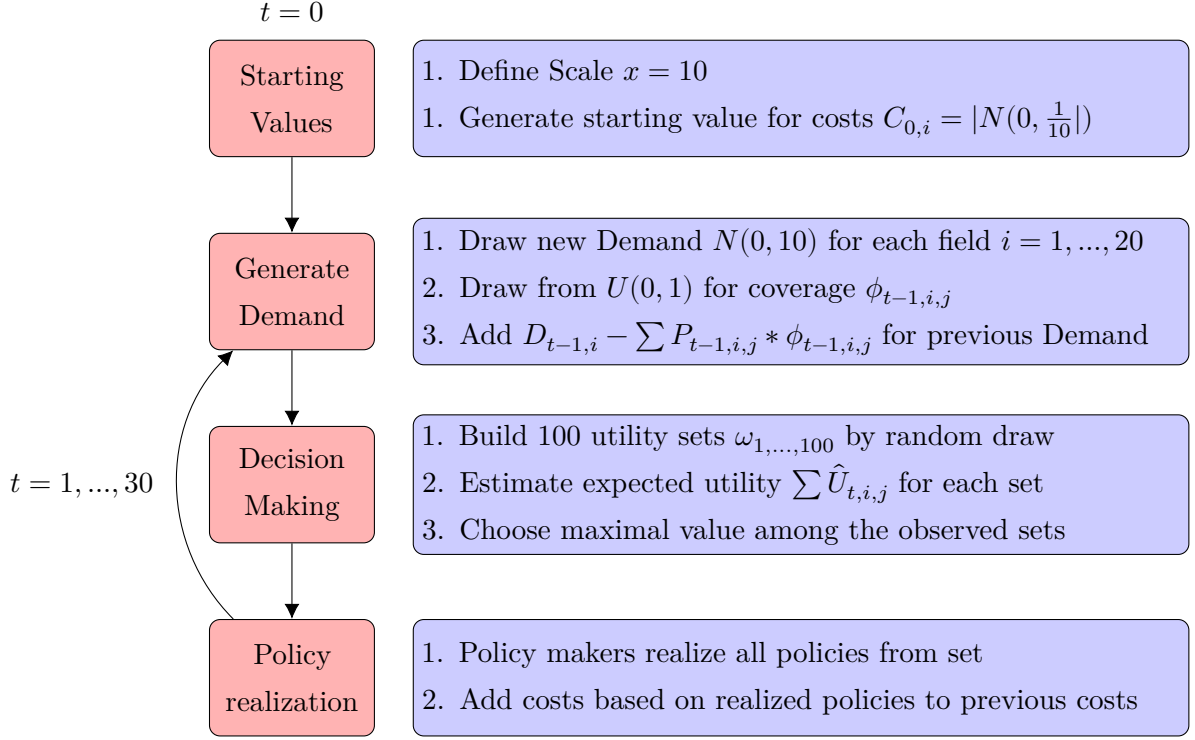


Figure 2: Simulation pseudo code.

The simulation generates a dataset that includes the number of policies fulfilled annually within each field and the corresponding functional costs for each period. Utilizing this, I compute the percentage changes in output for each field and period. Running the model under those conditions produces punctuated outputs, as seen in the Supplementary Information (SI). I employ quantile regression to visually represent functional costs' influence on policy punctuations. This method enables me to predict the impact of functional costs across various quantiles (τ) of percentage changes. If functional costs influence policy punctuations, I anticipate observing a negative coefficient of functional costs for high values of τ (see also [Breunig and Jones 2011](#)). To ensure representative results, I repeat this procedure 100 times. Figure 3 showcases the resulting coefficients derived from these 100 simulations.

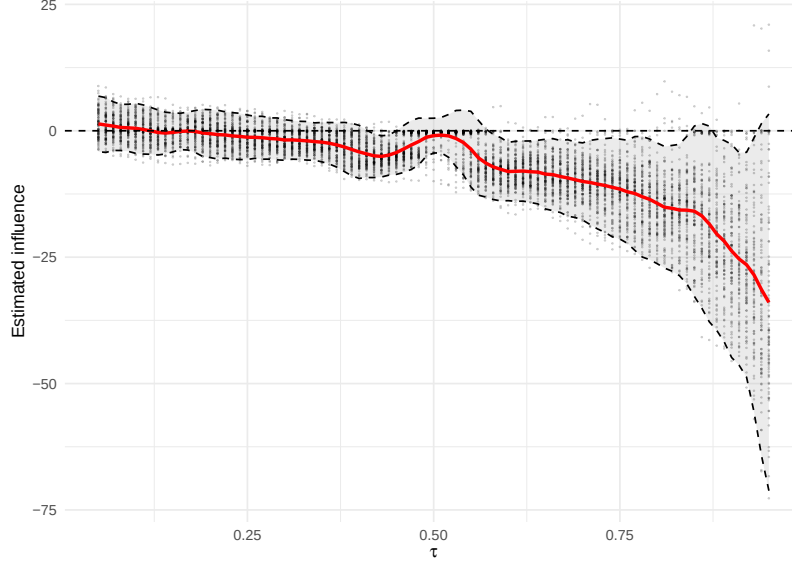


Figure 3: Results of quantile regression based on simulated scenarios. The simulation was conducted 100 times. The dependent variable are percentage changes per field and period; the independent variable is period standardized costs per field similar to the specification outlined in Equation 12. The quantile regression includes a time trend control to capture demand dynamics, similar to yearly fixed effects as in Equation 12. Each dot represents one coefficient estimate from a simulated scenario. The main line represents the mean coefficient estimate across all scenarios. The upper and lower lines mark the 95% confidence interval.

It becomes evident that higher quantile changes are inversely correlated with the functional cost factor under this model. The connection is especially prevalent in higher quantiles, thus resulting in more extreme policy changes. This suggests that greater costs in a field during a specific time period lead to less severe policy punctuations in the model. Conversely, lower relative costs exacerbate the severity of policy shifts. This can be summarized into one expectations for the empirical setting:

- E: The higher|lower the (relative) subsystem functional costs in a given period, the less|more pronounced are policy punctuations

5 Empirical strategy

The model shows that under certain circumstances functional costs can influence the occurrence of policy punctuations. Yet, the question remains if this pattern also manifests in

reality. To answer this question I will use legislative production of the European Union to test the expectations formulated above.

5.1 Case selection

The European Union (EU) is selected as the focus of this study primarily because its complex legal and institutional structure provides an ideal environment to test the theoretical argument about policy punctuations and dynamic feedback mechanisms. The EU's diversity in policy fields and decision-making processes is crucial for examining how variations across sectors influence these mechanisms. Its relative autonomy from direct public pressure, unlike national governments, allows for the analysis of functional costs and dynamic feedback on policy outcomes in a controlled environment, less affected from the confounding influence of public opinion ([Princen 2013](#)).

Moreover, the EU's extensive legal documentation, with comprehensive data available from 1980 to 2015 across various policy fields, supports a detailed longitudinal analysis. The use of the EuroVoc classification system structures legislative acts into distinct fields, providing a systematic framework to measure and compare policy changes over time ([Kaplaner, Knill, and Steinebach 2025](#)). Additionally, the EU's interconnected legal landscape, as seen through its network of citations and references, facilitates the application of network analysis ([Fjelstul 2019](#)) to quantify subsystem costs, directly supporting the theoretical examination.

Lastly, I leverage the EU's reliance on three legal instruments — Regulations, Directives, and Decisions — and their variance in application ([Hurka and Steinebach 2021](#)) to test the theoretical argument. The choice of legal instrument significantly influences the relevance of functional costs and, consequently, the applicability of the proposed mechanism. Regulations typically incur higher upfront functional costs due to their need for detailed drafting and harmonization to ensure immediate and uniform application across all Member States. This places a substantial burden on EU institutions during the drafting phase as they work to align these policies with the broader legal and regulatory landscape. Directives, in contrast, shift the burden of implementation to Member States, which adapt them into their national legal systems, thereby externalizing part of the functional costs (see also [Haag, Hurka, and Kaplaner 2024](#)) which should effect policy making patterns (see also [Fernández-I-Marín et al. 2024](#)). Decisions, similarly, are likely to incur lower functional costs than

Regulations, as they target specific entities, sectors, or Member States, requiring minimal harmonization.

By exploiting this variation in functional costs across legal instruments, the study tests the theoretical argument’s applicability. The expectation is that the effect of functional costs on policy punctuations will be higher for Regulations compared to Directives or Decisions. This setup makes the EU an optimal case for testing the theoretical framework.

5.2 Dependent Variable

The dependent variable in this analysis is the *yearly percentage change in policy outputs* across the European Union (EU) from 1980 to 2015, captured based on 69,483 policy outputs distributed across 16 distinct fields. The outputs are taken from the Fjelstul (2019) dataset and are limited to secondary legislation. Additionally, I employ the EuroVoc classification system to structure policy outputs into distinct fields (see also Kaplaner, Knill, and Steinebach 2025). EuroVoc provides a standardized and comprehensive categorization of EU legislative acts, allowing for consistent measurement and field-specific comparisons. By using this system, the analysis ensures that sectoral distinctions are consistently applied. Measuring the yearly percentage changes captures the relative fluctuations in policy activity and allows for the identification of punctuations as deviations from periods of stasis or incremental change.

The calculation for percentage change follows the standard approach:

$$Pct.change_{i,l,t} = \frac{N_{i,l,t} - N_{i,l,t-1}}{N_{i,l,t-1}} * 100 \quad (9)$$

Where:

- $N_{i,l,t}$ is the number of policy outputs in field i , legislative instrument l , at time t

This operationalization captures the intensity of change within each field, providing a dynamic view of legislative activity. The approach normalizes variations by assessing percentage changes rather than absolute counts, allowing for cross-sectoral comparisons over time.

Lastly, as noted in Equation 9 the percentage changes are calculated separately for each legal instrument. As previously mentioned, the logic behind the choice of instrument varies

significantly ([Hurka and Steinebach 2021](#)), affecting the amount of functional costs incurred by the EU. Disaggregating the percentage changes by legal instrument allows testing of these effects. The observation period starts in 1980; however, the first observation year is dropped due to its use in calculating the percentage changes. Additionally, because a lagged version of this variable is used as a control (see Section 5.5), observations from 1981 are also excluded from the analysis. Finally, since not all periods across all fields and instruments had legislative output, I exclude periods that follow a period with zero output to avoid calculations resulting in infinite values. For a detailed overview of the number of included periods per policy field, please refer to the supplementary information (SI).

5.3 Independent variable

I assess functional costs through the average degree of the legal citation network within a given field at a given time. This approach is grounded in network theory, where the degree of a node indicates the number of connections it has with other nodes. In the context of legislative outputs, the degree represents the extent to which a given policy item cites or is cited by other policy items. The average degree, in turn, captures, on average, how interconnected the nodes within the network are.

This metric reflects the complexity and interconnectedness (see [Bonchev and Buck 2005](#)) of the legal framework within each field and serves as a proxy for the subsystem’s functional costs. The reasoning is that the more interconnected and interdependent the legislative items are, the higher the cost for policy-makers to create, adapt, or implement new policies. Each additional connection implies that any new policy must consider, align with, or adjust to a more significant number of existing regulations, thereby increasing the drafting, harmonization, and compliance efforts. These efforts translate into higher functional costs, as policy-makers need to ensure consistency, address conflicts, and integrate new measures into an increasingly complex and interconnected legislative environment ([Burgin and Mestdagh 2020](#); [Koniaris, Anagnostopoulos, and Vassiliou 2018](#)).

Additionally, a higher average degree signifies a dense regulatory network, suggesting that changes in one part of the subsystem could lead to ripple effects throughout the network, further complicating the policy-making process. This interconnectedness requires additional resources, expertise, and time, further raising the functional costs of policy-making. The average degree encapsulates the effort and resources needed to navigate and modify the

subsystem’s legal landscape, reflecting how costly it is to implement changes that align with the existing regulatory framework (Kütt and Kask 2021; Burgin and Mestdagh 2020).

Similar approaches have been used by other scholars either to (partially) assess the complexity of individual provisions (Katz and Bommarito II 2014; Hurka, Haag, and Kaplaner 2022; Senninger 2023) or to assess the complexity of the legal landscape as a whole (Lee, Lee, and Yang 2019). The average degree $D_{i,t}$ for a field i at time t is defined as:

$$D_{i,t} = \frac{\sum_j^{N_{t,i}} k_{j,i,t}}{N_{j,i,t}} \quad (10)$$

Here, $N_{t,i}$ represents the total number of policy items in field i at time t , while $k_{j,i,t}$ denotes the degree of policy item j . Again, I use the EuroVoc classification for field identification. Additionally, I include connections where at least one node belongs to the specified field. The data on the connections between legal documents is taken from Fjelstul (2019). Only active and legally binding policy items are included for each time period to ensure that the measure accurately reflects the current functional environment. This information is taken from Borrett and Laurer (2020). This ensures the analysis focuses on the actual subsystem costs impacting policy-making within each period rather than including outdated or repealed documents.



Figure 4: Avg. Degree over time across all policy fields.

Yet, some caveats arise when using this measure. As shown in Figure 4, the average degree of all policy fields exhibits an upward trend over time (see also Hurka, Haag, and Kaplaner 2022). This trend suggests that using the average degree directly may conflate the impact of rising costs with a general time trend rather than isolating the specific effect of functional costs in the legislative process.

I use a scaled version of the average degree (*SAD*) to address this issue. According to the theoretical model, policy-makers operate within specific time frames when deciding on policy outputs, making a field’s *relative* costs more important than absolute values. Therefore, to better align the measure with this theoretical perspective and eliminate the risk of capturing mere temporal trends, I scale the average degree relative to the costs observed in other policy fields within the same period.

This adjustment achieves two main objectives: first, it removes the time trend from the data, as demonstrated in Figure 3, ensuring that the measure focuses on cross-sectional variance rather than temporal patterns. Second, it aligns with the theory that policy-makers prioritize relative costs, making the measure more theoretically consistent and empirically robust.

The final measure can be expressed as:

$$SAD_{i,t} = \text{Scaled Avg. Degree}_{i,t} = \frac{D_{i,t} - \mu_t}{\sigma_t} \quad (11)$$

Where $D_{i,t}$ is the average degree for field i , and μ_t is the mean of all policy fields within the same period with σ_t representing the standard deviation. This formulation ensures that the measure reflects the relative costs within the policy landscape while accounting for temporal dynamics.

Additionally, in the models outlined in Section 5.5, the variable is lagged by one period (SAD_{t-1}) to account for the fact that policy-makers react to the information available from the previous period when making decisions. This lagged structure reflects the temporal nature of decision-making processes. It aligns with the theoretical model, which assumes policy-makers respond to the latest data rather than contemporaneous conditions.

5.4 Controls

The model includes several controls to refine the analysis. First, yearly fixed effects account for time-specific factors (e.g., economic cycles or political shifts), isolating the relationship between functional costs and policy changes from broader temporal fluctuations. Field fixed effects capture differences across policy areas, ensuring that comparisons of functional costs are made within each field’s unique context. Controls for legal instruments (Regulations, Directives, Decisions) adjust for their distinct administrative implications. Lastly, a lagged percentage change accounts for temporal dependencies in policy activity, helping to capture

autocorrelations and reduce the influence of past fluctuations on current outcomes (see also [Holyoke and Brown 2019](#)).

5.5 Model specification

The following quantile regression model is used to estimate the conditional quantiles of the percentage change in policy outputs:

$$\begin{aligned}
Q_{Pct.change_{i,l,t}}(\tau|X_{i,t}) = & \alpha_\tau + \beta_{1,\tau} \cdot SAD_{i,t-1} + \beta_{2,\tau} \cdot Pct.change_{i,l,t-1} + \\
& \gamma_\tau^{Regulation} \cdot Regulation_i + \gamma_\tau^{Directive} \cdot Directive_i + \\
& \sum_i \lambda_\tau^{Field_i} \cdot Field_i + \sum_t \mu_\tau^{Year_t} \cdot Year_t
\end{aligned} \tag{12}$$

Where:

- $Q_{Pct.change_{i,l,t}}(\tau|X_{i,t})$ denotes the conditional quantile function of the percentage change in policy outputs for field i , legislative instrument l , and time t at quantile τ .
- $SAD_{i,t-1}$ represents the scaled average degree for field i in the previous period.
- $Pct.change_{i,l,t-1}$ is the lagged percentage change in outputs for field i instrument l at time $t - 1$.
- $Regulation_i$ and $Directive_i$ are dummy variables indicating whether the Pct. Changes stem from Regulations or Directives. Decisions serve as the baseline category.
- $\sum_i \lambda_\tau^{Field_i} \cdot Field_i$ represents the sum of dummy variables for different policy fields, with one field as the baseline.
- $\sum_t \mu_\tau^{Year_t} \cdot Year_t$ represents the sum of dummy variables for different years, with one year as the baseline.

The model is estimated using *conditional quantile regression*, which allows for assessing the influence of the independent variables across different quantiles (τ) of the percentage change distribution ([Koenker and Bassett 1978](#)).

The coefficient $\beta_{1,\tau}$ represents the effect of the SAD_{t-1} on the τ -th quantile of the percentage change in policy outputs. Specifically, it indicates how strong the costs of a field's legal network are associated with the percentage change in policy outputs at different points in the conditional distribution.

A positive $\beta_{1,\tau}$ would suggest that — for that specific quantile τ — an increase in the SAD_{t-1} (representing higher relative subsystem costs) is associated with an increase in the percentage change of policy outputs. Conversely, a negative $\beta_{1,\tau}$ would indicate that higher costs in the previous period are associated with a decrease in the percentage change for that quantile.

This interpretation allows an understanding of how the impact of functional cost varies across different policy activity levels, e.g., whether it has a more substantial effect at higher or lower distribution quantiles. This is especially useful in studying policy punctuations since the interest is primarily on the impact in the highest quantiles; therefore, in the cases of punctuated policy change (see [Breunig 2011](#); [Breunig and Jones 2011](#)).

In addition to the primary model, a secondary model is estimated where SAD_{t-1} interacts with the type of legal instrument. This interaction allows for isolating the exact mechanism by which the effect of functional costs varies based on the specific legislative instrument used — given the variation in the relevance of the functional costs associated with each, as outlined earlier. The quantile regression model for this interaction is specified as follows:

$$\begin{aligned}
Q_{Pct.change_{s,l,t}}(\tau|X_{s,t}) = & \alpha_\tau + \beta_{1,\tau} \cdot SAD_{s,t-1} + \beta_{2,\tau} \cdot Pct.change_{s,l,t-1} \\
& + \gamma_\tau^{Regulation} \cdot Regulation_s + \gamma_\tau^{Directive} \cdot Directive_s \\
& + \delta_\tau^{Regulation} \cdot (SAD_{s,t-1} \cdot Regulation_s) + \delta_\tau^{Directive} \cdot (SAD_{s,t-1} \cdot Directive_s) \\
& + \sum_i \lambda_\tau^{Field_i} \cdot Field_i + \sum_t \mu_\tau^{Year_t} \cdot Year_t
\end{aligned} \tag{13}$$

The inclusion of the interaction terms $\delta_\tau^{Regulation}$ and $\delta_\tau^{Directive}$ between $SAD_{i,t-1}$ and the legal instruments (Regulation and Directive) allows the model to isolate how the impact of subsystem functional costs (as measured by $SAD_{i,t-1}$) varies depending on the type of legislative instrument. Decisions serve as the baseline category.

This interaction effect is crucial, as it reflects the theoretical expectation that the functional costs associated with each instrument influence policy changes differently. The coefficients $\delta_\tau^{Regulation}$ and $\delta_\tau^{Directive}$ capture how the effect of subsystem functional costs changes when considering Pct. Changes in Regulations or Directives, compared to Decisions.

If the interaction coefficient $\delta_\tau^{Regulation}$ is positive, it would indicate that the effect of functional costs on the percentage change in policy outputs is amplified in the case of Regulations, consistent with the higher functional costs they impose. For Directives, if $\delta_\tau^{Directive}$ is

smaller or even negative, it would suggest a diminished or reversed effect, as the relevance of functional costs in Directives is theoretically lower. Decisions, as the baseline, are implicitly compared through these interaction terms.

Estimating this secondary model allows for testing the exact mechanism proposed in the theory. It reveals if the variations may not be solely tied to field-specific factors; rather, the results could show direct influence moderated by the degree to which functional costs are relevant to the chosen legal instrument.

The estimation is performed using the R package *quantreg* (Koenker 1999) employing the *Frisch-Newton approach*, which is efficient for quantile regression models, especially when dealing with large data sets. For the standard errors, I rely on Huber sandwich estimates as implemented in *quantreg*.

6 Results

6.1 Results from quantile regression

The results of the quantile regression are visualized in Figure 5, displaying the coefficient estimates from the specification in Equation 12 alongside their respective 95 percent confidence intervals. I will focus on the upper end of the distribution, specifically the 90th and 95th quantiles, as they capture cases of significant policy change. At this point, it is essential to note that overall, the patterns of the change distributions follow a punctuated pattern as assessed in the SI.

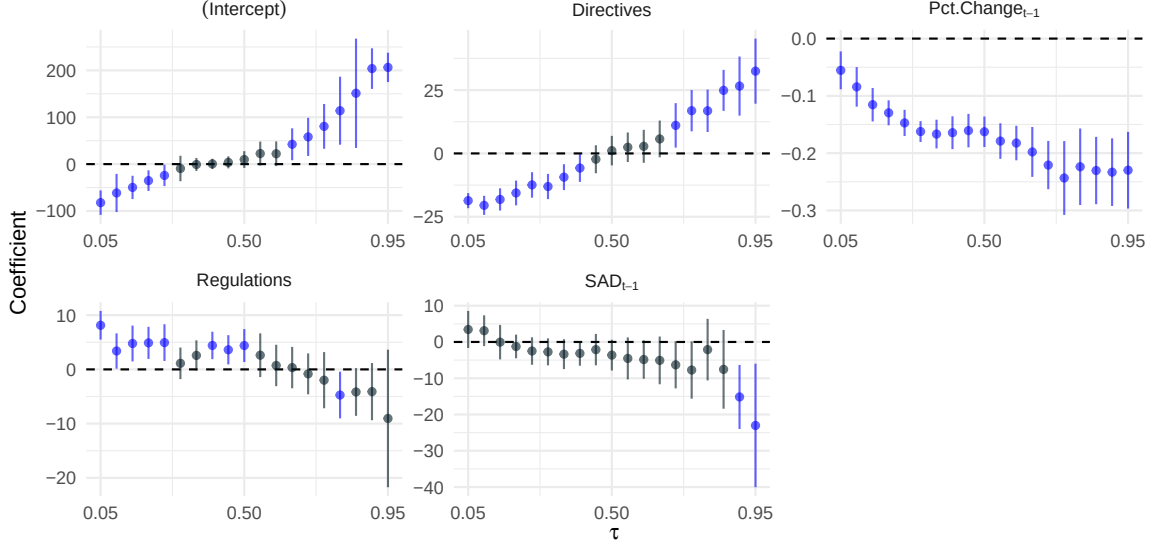


Figure 5: Coefficients retrieved from quantile regression across multiple values of τ for the specification outlined in Equation 12. The lines mark the respective 95% confidence interval. Coefficients that are (at least) significant on the 5% level are marked in blue.

Focusing on the main independent variable SAD_{t-1} , the analysis reveals a strong negative effect at the 90th ($\tau = .9$) and 95th quantiles ($\tau = .95$), statistically significant ($p < 0.01$). In the framework of quantile regression, this indicates that at these upper quantiles of the conditional distribution of percentage change, higher relative functional costs are associated with more minor percentage changes in policy outputs. In essence, relatively high subsystem costs reduce the likelihood or magnitude of extreme policy changes at the highest quantiles.

The coefficient size for SAD_{t-1} is -15 for $\tau = .9$ and -23 for $\tau = .95$. These values suggest that, for every one-unit increase in SAD_{t-1} — which can be interpreted as an increase of the magnitude of one standard deviation — the percentage change in policy outputs at the 90th and 95th quantiles decreases by -15 and -23 percentage points respectively. This sizable coefficient suggests a substantial dampening effect of high functional costs within a policy field on extreme policy changes, highlighting the stabilizing influence high costs might induce.

Importantly, since SAD_{t-1} can also take on negative values (indicating relatively low subsystem costs), the model predicts that, in such cases, there is an increased likelihood of policy punctuations at the upper quantiles. When functional costs within a subsystem are low,

the constraint on policy adjustments is minimized, making it more likely that large, punctuated changes occur. This pattern highlights the dual nature of subsystem costs: while high costs act as a stabilizing force, reducing the probability of extreme policy shifts, low costs facilitate these significant changes, thus increasing the likelihood of policy punctuations.

This interpretation demonstrates that subsystem costs have a significant and directional influence on the distributional shape of policy changes. It shows that low costs create a wider upper tail, indicating an environment conducive to large-scale policy shifts, while high costs compress this tail, dampening the extent of possible punctuations. This finding aligns with the theoretical expectation derived from the theoretical model that the amount of functional costs — whether high or low— modulate the likelihood and scale of extreme policy changes, thereby shaping the dynamics of policy punctuations systematically.

6.2 Mechanism check

But can these effects really be attributed to the higher-cost environment? I also test the interaction between the costs and the respective legal instruments to answer this question. In these models, Decision serves as the baseline category. Figure 6 shows the resulting coefficient estimate of this specification with their respective 95% confidence intervals.

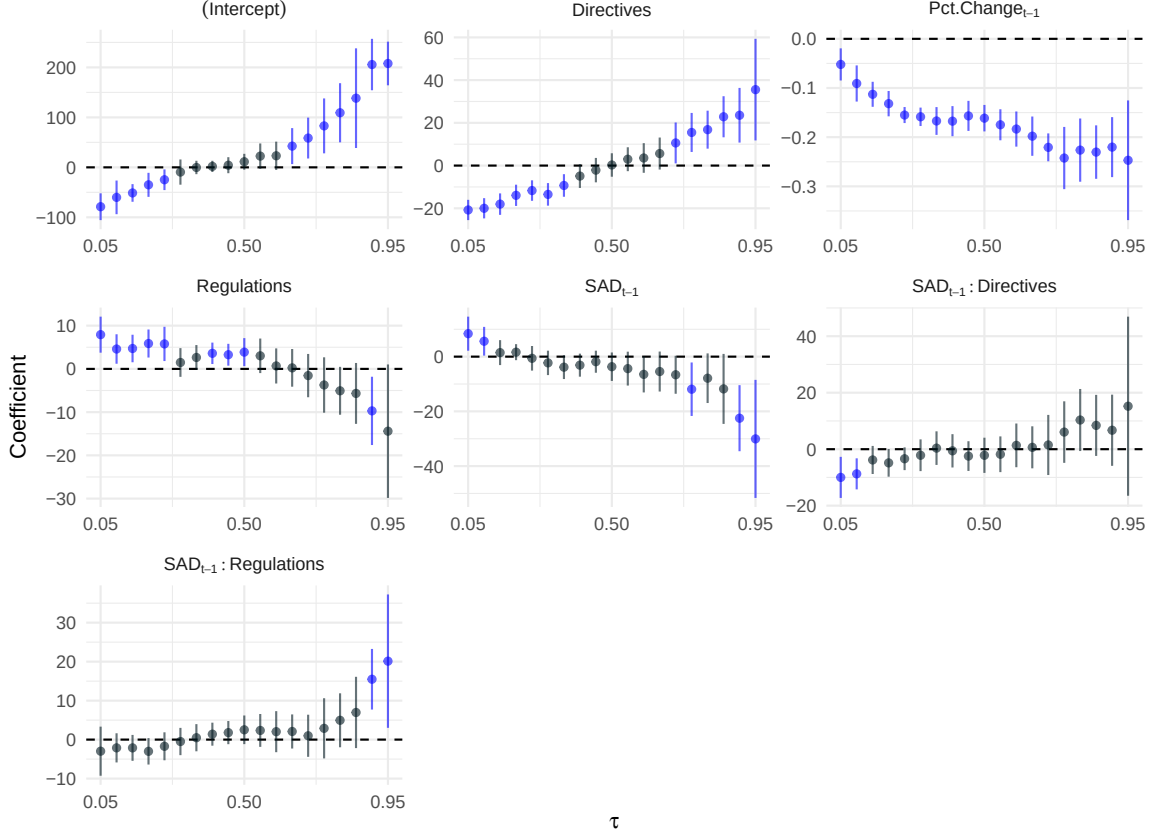


Figure 6: Coefficients retrieved from quantile regression across multiple values of τ for the specification outlined in Equation 13. The lines mark the respective 95% confidence interval. Coefficients that are (at least) significant on the 5% level are marked in blue.

Consistent with the earlier results, the model suggests a strong negative effect for the upper tails of the distribution for SAD_{t-1} . Turning to the interaction effects, the model shows a strong and statistically significant ($p < 0.05$) positive interaction between Regulations and SAD_{t-1} . This indicates that, as expected, the effect of costs on percentage changes is significantly higher for Regulations when compared to the baseline (Decisions). This result aligns with the theoretical argument that the greater complexity and harmonization requirements associated with Regulations amplify the impact of functional costs on policy outputs, thereby dampening the likelihood of large policy punctuations at the upper quantiles.

Notably, the interaction term for Directives shows no statistically significant effect, suggesting that the impact of costs on percentage changes does not differ from the baseline (Decisions). This finding is consistent with the notion that Directives externalize much of

the functional costs to Member States, resulting in lower immediate functional costs for the EU. As a result, the influence of subsystem complexity on the magnitude of policy changes remains similar to that of Decisions, which also typically incur lower costs.

The differential effects across legal instruments provide strong empirical support for the mechanism by demonstrating a pattern that aligns with theoretical expectations. When functional costs are high, they constrain significant policy changes in Regulations, which demand greater resources and effort in navigating functional costs. In contrast, in Directives and Decisions, where the susceptibility to functional costs is lower or externalized to Member States, the constraining effect is weaker or absent.

This variation across legal instruments reduces the likelihood that the observed relationship is due to a spurious correlation. If the relationship between functional costs and policy changes were spurious, I would expect the strength of the effect to remain relatively constant across the legal instrument. Instead, the pattern observed — where the constraining effect of costs is strongest in Regulations and weaker in Directives and Decisions — follows the theoretical considerations about their susceptibility to these costs. The alignment between theory and empirical results strengthens the credibility of the proposed mechanism, suggesting that functional costs are a crucial factor in shaping the scale and likelihood of policy punctuations.

7 Discussion

This study highlights the critical role of functional costs in shaping policy punctuations within complex legislative systems, with the European Union (EU) serving as a case study. The findings provide strong empirical support for the theoretical framework linking subsystem complexity, functional costs, and patterns of policy change. Specifically, functional costs—measured through the legal network’s average degree—significantly influence policy output distributions, especially at the upper quantiles where extreme policy changes are observed.

A key insight is that the impact of functional costs varies with different types of legal instruments. Regulations, which entail high upfront costs due to the need for harmonization and compliance across member states, reduce the likelihood and magnitude of significant policy shifts. This supports the idea that high-cost environments promote policy stability.

In contrast, Directives and Decisions, with lower or more externalized costs, are less sensitive to these constraints, making them more prone to policy punctuations. This finding underscores how the nature of legal instruments interacts with functional costs to shape policy dynamics.

7.1 Practical Implications

These findings have important practical implications. Recognizing that functional costs are both influential and adaptable offers policy-makers a valuable lever for reform. By implementing institutional reforms, fostering innovation, or simplifying legislative processes, governments can reduce these costs over time. Although such strategies may require upfront investments and may not yield immediate results, the long-term benefits include reducing policy inertia and enabling more agile legislative action. Efforts to streamline drafting and compliance processes can prevent legislative stagnation. Early examples of this approach can already be seen in the EU's REFIT program, which aims to simplify legislation and reduce unnecessary burdens.

7.2 Limitations and Future Research

While these findings provide robust insights within the EU context, they may have limitations when generalized to other political systems. The EU's unique supranational structure, relative autonomy from direct electoral pressures, and complex regulatory framework differ significantly from national governments, where electoral cycles and public opinion often play a more direct role in shaping policy. However, the theoretical framework presented here could be adapted to other governance systems, as all policy-makers must navigate existing legal frameworks when developing new policies. The focus on functional costs and subsystem complexity offers a valuable foundation for exploring policy dynamics across diverse political environments.

One limitation of this study is the operationalization of functional costs through the average degree of the legal network, which may not capture all dimensions of these costs. While this metric reflects the complexity and interconnectedness of the legal subsystem, it may overlook factors such as administrative burdens, legal intricacies, or the actual resource expenditures involved in policy-making. Additionally, the assumption that network degree directly correlates with functional costs may not always hold, as some highly interconnected policy areas could mitigate additional costs through standardized procedures.

Moreover, measuring policy punctuations solely in terms of percentage changes might not fully capture the significance or substantive impact of these shifts. Percentage changes indicate the magnitude of change relative to a prior state but may not reflect the qualitative importance of those changes. Some policy developments may have far-reaching consequences despite only modest quantitative shifts, while large changes in less critical areas may have minimal overall impact. Future research could benefit from incorporating qualitative analyses and assessments of policy impact to provide a more comprehensive understanding of policy dynamics.

7.3 Conclusion

This paper introduces the concept of dynamic feedback mechanisms in complex legislative environments, bridging Punctuated Equilibrium Theory with insights from complex systems and information processing theories. It emphasizes the role of subsystem complexity in determining the rate of policy change and demonstrates how functional costs can drive both policy stability and change.

By illuminating the dynamic interplay between functional costs, subsystem complexity, and policy change, this study offers a nuanced understanding of policy punctuations. It highlights the importance of addressing structural constraints while recognizing opportunities for innovation and reform. These insights provide valuable perspectives for policy-makers navigating the challenges of complex legislative systems and seeking to balance stability with responsiveness in policy evolution.

References

- Adam, Christian, Steffen Hurka, Christoph Knill, and Yves Steinebach. 2019. *Policy Accumulation and the Democratic Responsiveness Trap*. 1st ed. Cambridge University Press. <https://doi.org/10.1017/9781108646888>.
- Baumgartner, Frank R., Christian Breunig, Christoffer Green-Pedersen, Bryan D. Jones, Peter B. Mortensen, Michiel Nuytemans, and Stefaan Walgrave. 2009. "Punctuated Equilibrium in Comparative Perspective." *American Journal of Political Science* 53 (3): 603–20. <https://doi.org/10.1111/j.1540-5907.2009.00389.x>.
- Baumgartner, Frank R., and Bryan D. Jones. 1993. *Agendas and Instability in American Politics*. Chicago: University of Chicago Press.
- Béland, Daniel. 2009. "Ideas, Institutions, and Policy Change." *Journal of European Public Policy* 16 (5): 701–18. <https://doi.org/10.1080/13501760902983382>.
- Bendor, Jonathan, Terry M. Moe, and Kenneth W. Shotts. 2001. "Recycling the Garbage Can: An Assessment of the Research Program." *American Political Science Review* 95 (1): 169–90. <https://doi.org/10.1017/S0003055401000041>.
- Birkland, Thomas A. 2006. *Lessons of Disaster: Policy Change After Catastrophic Events*. Washington, D.C.: Georgetown University Press.
- Bonchev, Danail, and Gregory A. Buck. 2005. "Quantitative Measures of Network Complexity." In *Complexity in Chemistry, Biology, and Ecology*, edited by Danail Bonchev and Dennis H. Rouvray, 191–235. Boston, MA: Springer US. https://doi.org/10.1007/0-387-25871-X_5.
- Borrett, Camille, and Moritz Laurer. 2020. "The CEPS EurLex Dataset: 142.036 EU Laws from 1952-2019 with Full Text and 22 Variables." Harvard Dataverse. <https://doi.org/10.7910/DVN/0EGYWY>.
- Breunig, Christian. 2011. "Reduction, Stasis, and Expansion of Budgets in Advanced Democracies." *Comparative Political Studies* 44 (8): 1060–88. <https://doi.org/10.1177/0010414011405169>.
- Breunig, Christian, and Bryan D. Jones. 2011. "Stochastic Process Methods with an Application to Budgetary Data." *Political Analysis* 19 (1): 103–17. <https://doi.org/10.1093/pan/mpq038>.
- Buchanan, James M., and Gordon Tullock. 1965. *The Calculus of Consent: Logical Foundations of Constitutional Democracy*. 19. Nachdr. Ann Arbor, Mich: Univ. of Michigan Press.
- Burgin, Mark, and Kees (C.N. J.) De Vey Mestdagh. 2020. "Complexity of Legal Pro-

- cesses and Systems.” In *IS4SI 2019 Summit*, 17. MDPI. <https://doi.org/10.3390/proceedings2020047017>.
- Cairney, Paul. 2016. *The Politics of Evidence-Based Policy Making*. London: Palgrave Macmillan.
- Desmarais, Bruce A. 2019. “Punctuated Equilibrium or Incrementalism in Policymaking: What We Can and Cannot Learn from the Distribution of Policy Changes.” *Research & Politics* 6 (3): 205316801987139. <https://doi.org/10.1177/2053168019871399>.
- Fernández-I-Marín, Xavier, Christoph Knill, Christina Steinbacher, and Yves Steinebach. 2024. “Bureaucratic Quality and the Gap Between Implementation Burden and Administrative Capacities.” *American Political Science Review* 118 (3): 1240–60. <https://doi.org/10.1017/S0003055423001090>.
- Fjelstul, Joshua C. 2019. “The Evolution of European Union Law: A New Data Set on the *Acquis Communautaire*.” *European Union Politics* 20 (4): 670–91. <https://doi.org/10.1177/1465116519842947>.
- Gould, Stephen Jay, and Niles Eldredge. 1977. “Punctuated Equilibria: The Tempo and Mode of Evolution Reconsidered.” *Paleobiology* 3 (2): 115–51. <https://doi.org/10.1017/S0094837300005224>.
- Haag, Maximilian, Steffen Hurka, and Constantin Kaplaner. 2024. “Policy Complexity and Implementation Performance in the European Union.” *Regulation & Governance*, January, rego.12580. <https://doi.org/10.1111/rego.12580>.
- Hall, Peter A. 1993. “Policy Paradigms, Social Learning, and the State: The Case of Economic Policymaking in Britain.” *Comparative Politics* 25 (3): 275. <https://doi.org/10.2307/422246>.
- Holyoke, Thomas T., and Heath Brown. 2019. “After the Punctuation: Competition, Uncertainty, and Convergent State Policy Change.” *State Politics & Policy Quarterly* 19 (1): 3–28. <https://doi.org/10.1177/1532440018788564>.
- Hurka, Steffen, and Maximilian Haag. 2020. “Policy Complexity and Legislative Duration in the European Union.” *European Union Politics* 21 (1): 87–108. <https://doi.org/10.1177/1465116519859431>.
- Hurka, Steffen, Maximilian Haag, and Constantin Kaplaner. 2022. “Policy Complexity in the European Union, 1993-Today: Introducing the EUPLEX Dataset.” *Journal of European Public Policy* 29 (9): 1512–27. <https://doi.org/10.1080/13501763.2021.1938174>.
- Hurka, Steffen, and Yves Steinebach. 2021. “Legal Instrument Choice in the European Union.” *JCMS: Journal of Common Market Studies* 59 (2): 278–96. <https://doi.org/10.1111/jcms.13068>.

- Jochim, Ashley E., and Peter J. May. 2010. "Beyond Subsystems: Policy Regimes and Governance: Jochim/May: Beyond Subsystems." *Policy Studies Journal* 38 (2): 303–27. <https://doi.org/10.1111/j.1541-0072.2010.00363.x>.
- Jones, Bryan D., and Frank R. Baumgartner. 2004. "A Model of Choice for Public Policy." *Journal of Public Administration Research and Theory* 15 (3): 325–51. <https://doi.org/10.1093/jopart/mui018>.
- . 2012. "From There to Here: Punctuated Equilibrium to the General Punctuation Thesis to a Theory of Government Information Processing: Jones/Baumgartner: Punctuated Equilibrium Theory." *Policy Studies Journal* 40 (1): 1–20. <https://doi.org/10.1111/j.1541-0072.2011.00431.x>.
- Jones, Bryan D., Frank R. Baumgartner, Christian Breunig, Christopher Wlezien, Stuart Soroka, Martial Foucault, Abel François, et al. 2009. "A General Empirical Law of Public Budgets: A Comparative Analysis." *American Journal of Political Science* 53 (4): 855–73. <https://doi.org/10.1111/j.1540-5907.2009.00405.x>.
- Jones, Bryan D., Tracy Sulkin, and Heather A. Larsen. 2003. "Policy Punctuations in American Political Institutions." *American Political Science Review* 97 (01): 151–69. <https://doi.org/10.1017/S0003055403000583>.
- Kaplaner, Constantin, Christoph Knill, and Yves Steinebach. 2025. "Policy Integration in the European Union: Mapping Patterns of Intersectoral Policy-Making over Time and Across Policy Sectors." *Journal of European Public Policy* 32 (1): 26–51. <https://doi.org/10.1080/13501763.2023.2288932>.
- Kaplaner, Constantin, and Yves Steinebach. 2022. "Why We Should Use the Gini Coefficient to Assess Punctuated Equilibrium Theory." *Political Analysis* 30 (3): 450–55. <https://doi.org/10.1017/pan.2021.25>.
- Katz, D. M., and M. Bommarito II. 2014. "Measuring the Complexity of the Law: The United States Code." *Artificial Intelligence and Law* 22 (4): 337–74.
- Kingdon, John W. 1995. *Agendas, Alternatives, and Public Policies*. 2nd ed. New York: Longman.
- Klijn, Erik-Hans, Bram Steijn, and Jurian Edelenbos. 2010. "THE IMPACT OF NETWORK MANAGEMENT ON OUTCOMES IN GOVERNANCE NETWORKS." *Public Administration* 88 (4): 1063–82. <https://doi.org/10.1111/j.1467-9299.2010.01826.x>.
- Koenker, Roger. 1999. "Quantreg: Quantile Regression." Comprehensive R Archive Network. <https://doi.org/10.32614/CRAN.package.quantreg>.
- Koenker, Roger, and Gilbert Bassett. 1978. "Regression Quantiles." *Econometrica* 46 (1): 33. <https://doi.org/10.2307/1913643>.

- Koniaris, Marios, Ioannis Anagnostopoulos, and Yannis Vassiliou. 2018. "Network Analysis in the Legal Domain: A Complex Model for European Union Legal Sources." *Journal of Complex Networks* 6 (2): 243–68. <https://doi.org/10.1093/comnet/cnx029>.
- Krutilla, Kerry. 2011. "Transaction Costs and Environmental Policy: An Assessment Framework and Literature Review." *International Review of Environmental and Resource Economics* 4 (3-4): 261–354. <https://doi.org/10.1561/101.000000035>.
- Kuhlmann, Johanna, and Jeroen van der Heijden. 2018. "What Is Known about Punctuated Equilibrium Theory? And What Does That Tell Us about the Construction, Validation, and Replication of Knowledge in the Policy Sciences?: What Is Known about Punctuated Equilibrium Theory?" *Review of Policy Research* 35 (2): 326–47. <https://doi.org/10.1111/ropr.12283>.
- Kütt, Andres, and Laura Kask. 2021. "Measuring Complexity of Legislation. A Systems Engineering Approach." In *Proceedings of Fifth International Congress on Information and Communication Technology*, edited by Xin-She Yang, R Simon Sherratt, Nilanjan Dey, and Amit Joshi, 1183:75–94. Singapore: Springer Singapore. https://doi.org/10.1007/978-981-15-5856-6_7.
- Lee, Bokwon, Kyu-Min Lee, and Jae-Suk Yang. 2019. "Network Structure Reveals Patterns of Legal Complexity in Human Society: The Case of the Constitutional Legal Network." Edited by Katarzyna Musial-Gabrys. *PLOS ONE* 14 (1): e0209844. <https://doi.org/10.1371/journal.pone.0209844>.
- Lorenz, M. O. 1905. "Methods of Measuring the Concentration of Wealth." *Publications of the American Statistical Association* 9 (70): 209. <https://doi.org/10.2307/2276207>.
- Mortensen, Peter B. 2009. "Political Attention and Public Spending in the United States." *Policy Studies Journal* 37 (3): 435–55. <https://doi.org/10.1111/j.1541-0072.2009.00322.x>.
- North, Douglass C. 1990a. *Institutions, Institutional Change, and Economic Performance*. The Political Economy of Institutions and Decisions. Cambridge ; New York: Cambridge University Press.
- . 1990b. "A Transaction Cost Theory of Politics." *Journal of Theoretical Politics* 2 (4): 355–67. <https://doi.org/10.1177/0951692890002004001>.
- Padgett, John F. 1980. "Bounded Rationality in Budgetary Research." *American Political Science Review* 74 (2): 354–72. <https://doi.org/10.2307/1960632>.
- Pierson, Paul. 2000. "Increasing Returns, Path Dependence, and the Study of Politics." *American Political Science Review* 94 (2): 251–67. <https://doi.org/10.2307/2586011>.
- . 2004. *Politics in Time: History, Institutions, and Social Analysis*. Princeton, NJ:

- Princeton Univ. Press.
- Princen, Sebastiaan. 2013. "Punctuated Equilibrium Theory and the European Union." *Journal of European Public Policy* 20 (6): 854–70. <https://doi.org/10.1080/13501763.2013.781822>.
- Rhodes, R. A. W. 1997. *Understanding Governance: Policy Networks, Governance, Reflexivity, and Accountability*. Public Policy and Management. Buckingham ; Philadelphia: Open University Press.
- Robinson, Scott E., Floun'say Caver, Kenneth J. Meier, and Laurence J. O'Toole. 2007. "Explaining Policy Punctuations: Bureaucratization and Budget Change." *American Journal of Political Science* 51 (1): 140–50. <https://doi.org/10.1111/j.1540-5907.2007.00242.x>.
- Sabatier, Paul A. 1988. "An Advocacy Coalition Framework of Policy Change and the Role of Policy-Oriented Learning Therein." *Policy Sciences* 21 (2-3): 129–68. <https://doi.org/10.1007/BF00136406>.
- Schmidt, Vivien A. 2010. "Taking Ideas and Discourse Seriously: Explaining Change Through Discursive Institutionalism as the Fourth 'New Institutionalism'." *European Political Science Review* 2 (01): 1. <https://doi.org/10.1017/S175577390999021X>.
- Schuck, P. H. 1992. "Legal Complexity: Some Causes, Consequences, and Cures." *Duke Law Journal* 42 (1): 1–52.
- Senninger, Roman. 2023. "What Makes Policy Complex?" *Political Science Research and Methods* 11 (4): 913–20. <https://doi.org/10.1017/psrm.2023.23>.
- Simon, Herbert A. 1957. *Models of Man: Social and Rational – Mathematical Essays on Rational Human Behavior in a Social Setting*. New York: Wiley.
- . 1962. "The Architecture of Complexity." *Proceedings of the American Philosophical Society* 106 (6): 467–82.
- Spiller, Pablo T., and Emerson H. Tiller. 1997. "Decision Costs and the Strategic Design of Administrative Process and Judicial Review." *The Journal of Legal Studies* 26 (2): 347–70. <https://doi.org/10.1086/467999>.
- Stauffer, Maxime, Isaak Mengesha, Konrad Seifert, Igor Krawczuk, Jens Fischer, and Giovanna Di Marzo Serugendo. 2022. "A Computational Turn in Policy Process Studies: Coevolving Network Dynamics of Policy Change." Edited by Ana Teixeira de Melo. *Complexity* 2022 (April): 1–17. <https://doi.org/10.1155/2022/8210732>.
- Sterman, John D. 2000. *Business Dynamics: Systems Thinking and Modeling for a Complex World*. International student edition. Boston Burr Ridge, IL Dubuque, IA Madison, WI New York San Francisco St. Louis Bangkok Bogotá Caracas Lisbon London Madrid

Mexico City Milan New Delhi Seoul Singapore Sydney Taipei Toronto: Irwin, McGraw-Hill.

Thelen, Kathleen. 2004. *How Institutions Evolve: The Political Economy of Skills in Germany, Britain, the United States, and Japan*. 1st ed. Cambridge University Press. <https://doi.org/10.1017/CBO9780511790997>.

Workman, Samuel, Bryan D. Jones, and Ashley E. Jochim. 2009. "Information Processing and Policy Dynamics." *Policy Studies Journal* 37 (1): 75–92. <https://doi.org/10.1111/j.1541-0072.2008.00296.x>.

Supplementary Information

A.1 Density curves and Lorenz curve for Pct.changes in simulation

Figure A.1 shows the distribution of pct. changes derived from an exemplary simulation of the model presented in the main text.

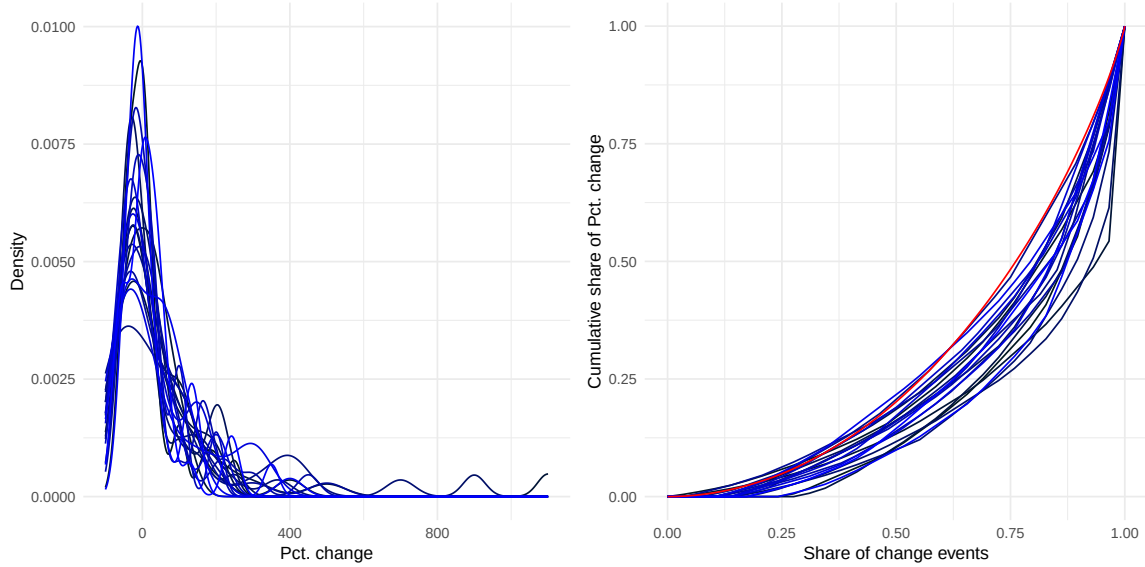


Figure A.1: Density and Lorenz curves of pct. changes of 20 simulated policy fields over a 30 year period

As the figure shows, all fields demonstrated punctuated patterns. To assess if the policy making across fields and instruments follows a punctuated pattern, I apply the method proposed by Kaplaner and Steinebach (2022). With the average Gini being $G = 0.551$ which implies a more punctuated distribution than a normal distribution ($G = .414$). Notably, this value lies close to the mean Gini value observed in the empirical distribution of EU legislative outputs (see below).

A.2 Density curves of Pct. changes by policy fields and legal instrument in the EU

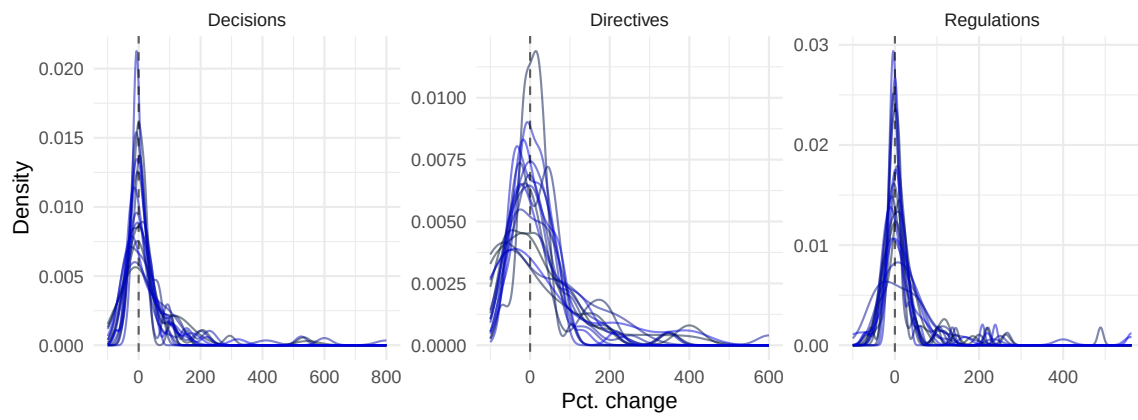


Figure A.2: Density curves of pct. changes across all policy fields split by legal instruments

A.3 Gini coefficient of Pct. changes by policy fields and legal instrument in the EU

Figure A.3 shows the Gini Coefficient based on the pct. changes of all policy fields split by legal instruments. The dotted line marks the Gini coefficient of a standard normal distribution (see above).

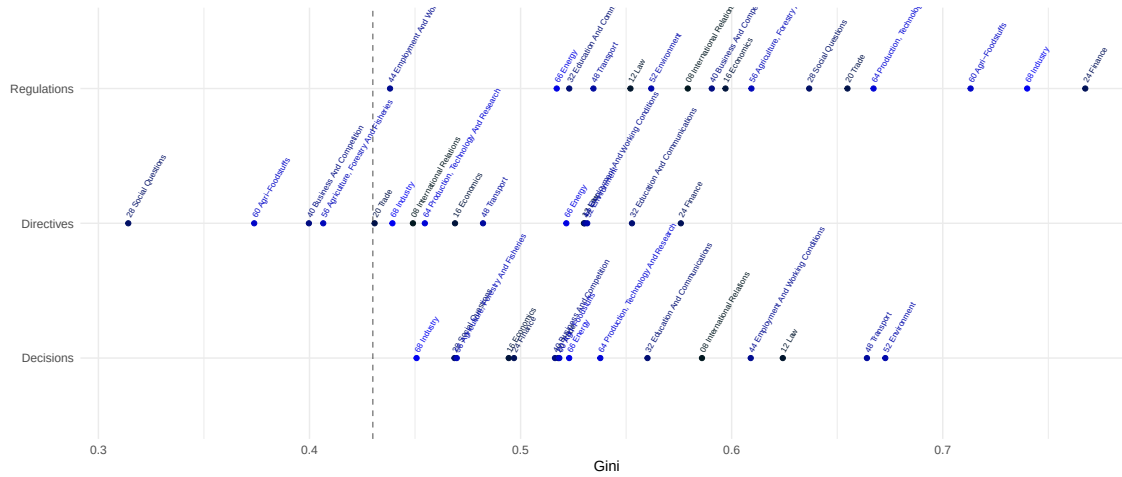
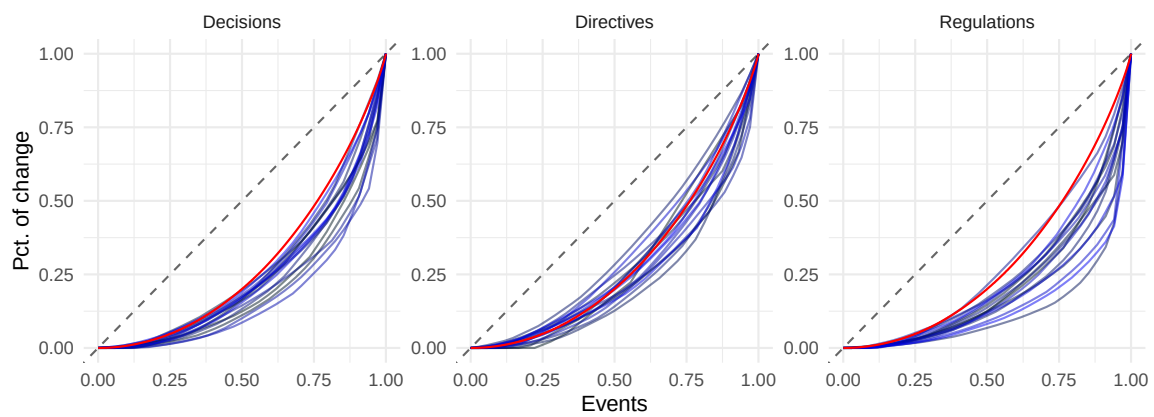


Figure A.3: Gini coefficient based on Pct. Changes per policy field and instrument, dotted line represents the Gini coefficient of a normal distribution. Values above that indicate a punctuated pattern (see [Kaplaner and Steinebach 2022](#)).

Looking at the graph, it is evident that nearly all fields across legal instrument have patterns that divert from a normal distribution.

A.4 Lorenz curves of Pct. changes by policy fields and legal instrument in the EU

The same pattern emerges when looking at the Lorenz-curve, that can visualize the inequality within distributions ([Lorenz 1905](#)). The red line indicates the Lorenz-curve of a normal distribution.



A.5 Main regression results table

Table A.1: Quantile regression results for selected quantiles.

	$\tau = .05$	$\tau = .5$	$\tau = .95$
(Intercept)	-82.23*** [-108.35, -56.12]	9.75 [-8.09, 27.59]	206.4*** [175.11, 237.69]
SAD (lagged)	3.45 [-1.66, 8.56]	-3.62* [-7.86, 0.62]	-23.02*** [-40.06, -5.99]
Regulations	8.14*** [5.49, 10.78]	4.4*** [1.36, 7.44]	-9.05 [-21.73, 3.64]
Directives	-18.64*** [-21.62, -15.66]	1.11 [-4.73, 6.94]	32.49*** [19.63, 45.34]
Pct. change (lagged)	-0.06*** [-0.09, -0.02]	-0.16*** [-0.19, -0.14]	-0.23*** [-0.3, -0.16]
Year	Yes	Yes	Yes
Field	Yes	Yes	Yes
N	1581	1581	1581
AIC	17116.35	17179.58	19874.81
BIC	17400.74	17463.97	20159.2

95% confidence intervals in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

A.6 Main regression results table with interaction

Table A.2: Quantile regression results for selected quantiles including interaction between SAD (lagged) and Legal instrument.

	$\tau = .05$	$\tau = .5$	$\tau = .95$
(Intercept)	-78.81*** [-105.61, -52.02]	11.21 [-4.69, 27.1]	207.84*** [164.14, 251.54]
SAD (lagged)	8.39*** [2.17, 14.62]	-3.68 [-8.88, 1.53]	-30.06*** [-51.63, -8.49]
SAD (lagged):Regulations	-2.97 [-9.27, 3.33]	2.51 [-1.16, 6.19]	20.13** [3.02, 37.23]
SAD (lagged):Directives	-9.99*** [-17.26, -2.71]	-2.16 [-8.39, 4.07]	15.22 [-16.49, 46.93]
Regulations	7.92*** [3.75, 12.08]	3.87** [0.63, 7.12]	-14.41* [-29.85, 1.03]
Directives	-20.78*** [-25.53, -16.04]	0.27 [-5.22, 5.76]	35.55*** [11.77, 59.32]
Pct. change (lagged)	-0.05*** [-0.08, -0.02]	-0.16*** [-0.19, -0.13]	-0.25*** [-0.37, -0.13]
Year	Yes	Yes	Yes
Field	Yes	Yes	Yes
N	1581	1581	1581
AIC	17103.4	17178.8	19862.05
BIC	17398.52	17473.92	20157.17

95% confidence intervals in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

A.7 Descriptive table main variables

Table A.3: Descriptive statistic main variables by policy field.

Field	N	Pct. Change at quantile:			Mean SAD
		5th	50th	95th	
08 International Relations	83	-64.1	1.64	145	-0.26
12 Law	93	-70.1	8.7	185	-0.909
16 Economics	89	-100	0	160	1.42
20 Trade	99	-29.7	1.71	60.3	0.767
24 Finance	98	-51.7	0	176	-0.637
28 Social Questions	99	-41.2	6.67	78.3	0.39
32 Education And Communications	99	-50	6.25	126	-0.954
40 Business And Competition	96	-68.2	0	175	1.81
44 Employment And Working Conditions	98	-57.9	0	154	-0.534
48 Transport	99	-59.2	4.17	145	-0.21
52 Environment	99	-55.6	3.03	157	0.236
56 Agriculture, Forestry And Fisheries	99	-39.6	1.07	62.1	0.732
60 Agri-Foodstuffs	99	-41.3	-0.411	111	0.206
64 Production, Technology And Research	99	-50	1.67	138	-0.843
66 Energy	91	-78.4	0	200	-1.23
68 Industry	99	-42.2	2.86	75.8	-0.0059

A.8 Robustness to different specification of independent variable

To assess the robustness of the model to the construction of the main independent variable, *SAD*, I apply an alternative transformation. Instead of scaling by period, I use ranked values. The results remain robust under this adjustment, demonstrating the model's stability. Only in the model including the interaction effect the significance level drops slightly for the 95th quantile, only being significant on the 10% level.

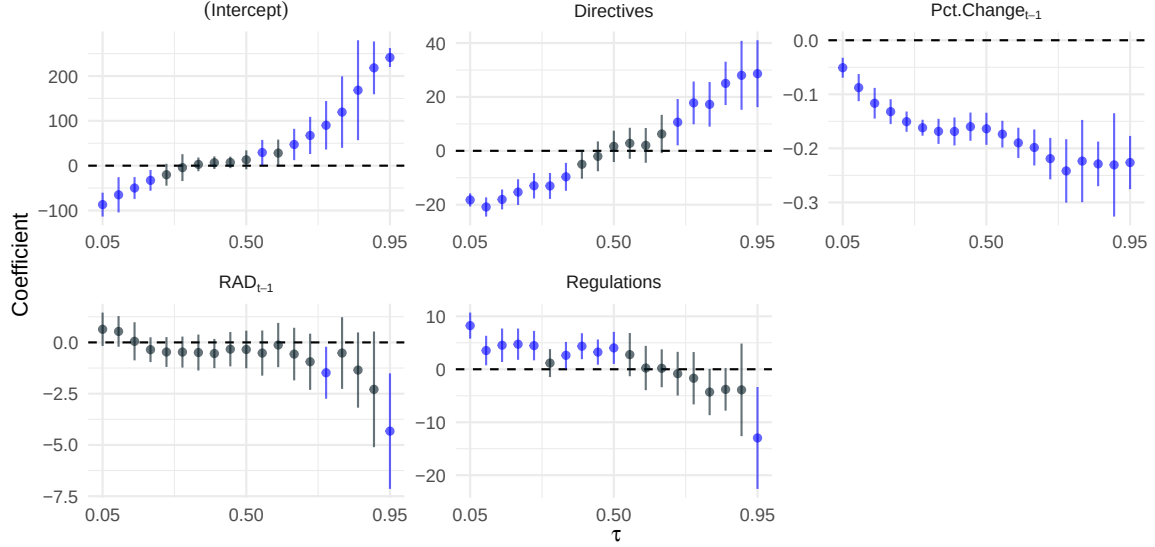


Figure A.4: Coefficient retrieved from quantile regression across multiple values of τ for the specification outlined in Equation 12 with different dependent variable (ranked, RAD_{t-1}). The lines mark the respective 95% confidence interval. Coefficient that are (at least) significant on the 5% level are marked in blue.

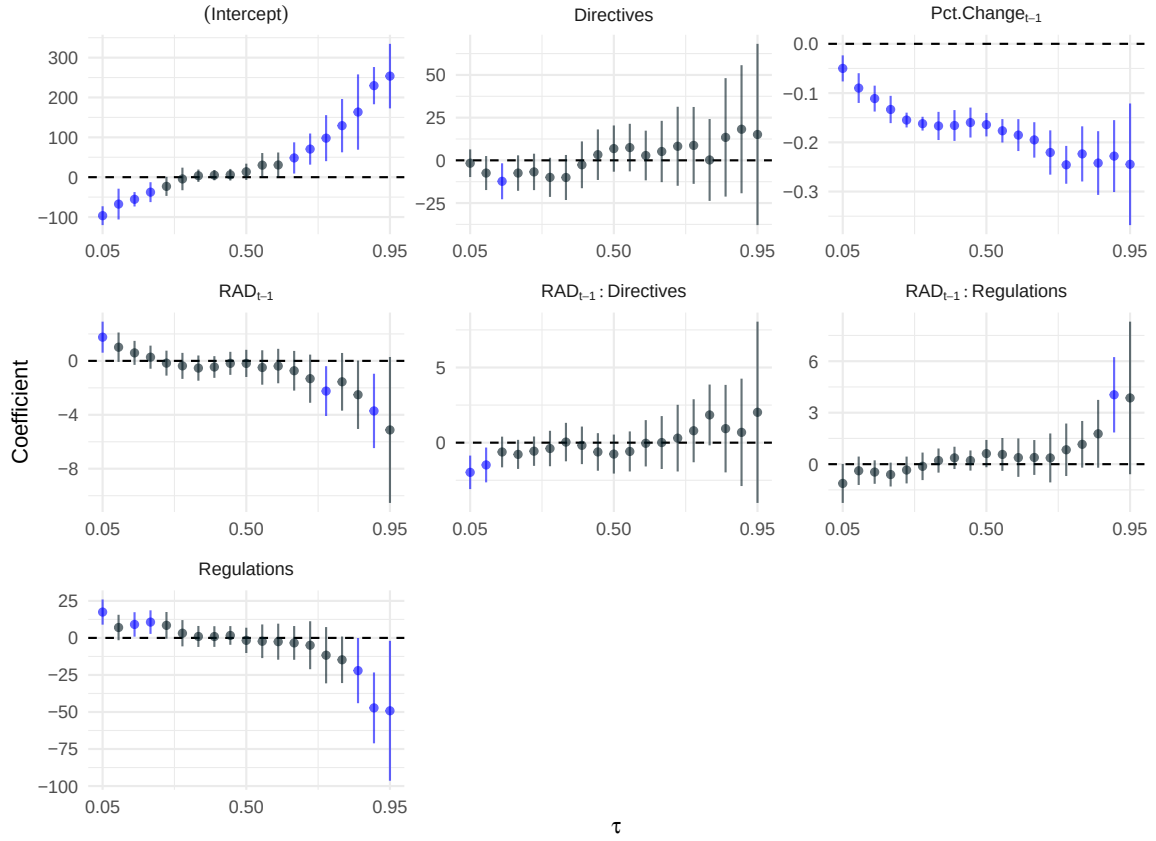


Figure A.5: Coefficient retrieved from quantile regression across multiple values of τ for the specification outlined in Equation 13 with different dependent variable (ranked, RAD_{t-1}). The lines mark the respective 95% confidence intervall. Coefficient that are (at least) significant on the 5% level are marked in blue.

A.9 Bivariate regression using main variable SAD

To evaluate potential suppression effects from the control variables, I estimate a simplified bivariate model that includes only the main independent variable, SAD , excluding all controls. As illustrated in Figure A.6, the effect persists even in this reduced model. The results indicate that the effect remains similar in magnitude to the main model (-20% compared to -23%) and is statistically significant at the 10% level.

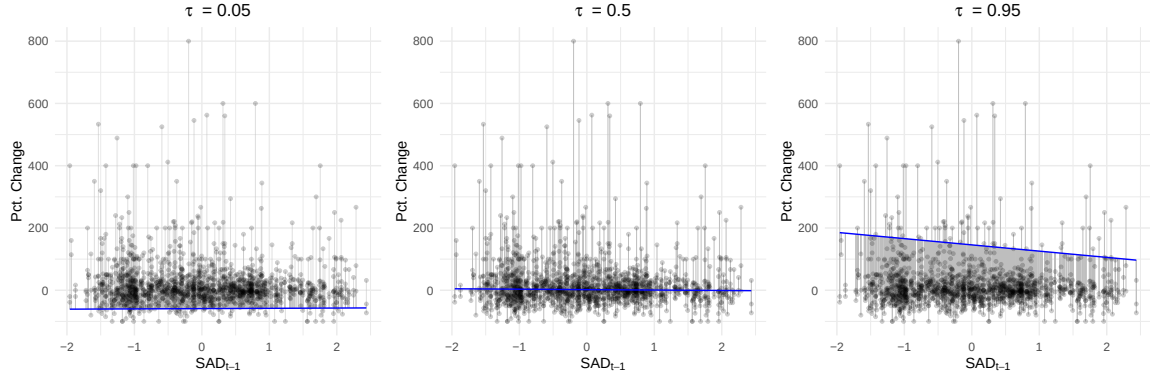


Figure A.6: SAD_{t-1} on x-axis, Pct. Change on y-axis. Blue line indicates regression line for different quantiles. Grey lines indicate distance from regression line.

Table A.4: Quantile regression results for selected quantiles for bivariate regression without controls.

	$\tau = .05$	$\tau = .5$	$\tau = .95$
(Intercept)	-58.94*** [-63.3, -54.57]	2.05** [0.33, 3.76]	145.73*** [127.35, 164.11]
SAD (lagged)	0.9 [-3.69, 5.49]	-1.38 [-3.02, 0.26]	-20.08* [-38.89, -1.27]
Year	Yes	Yes	Yes
Field	Yes	Yes	Yes
N	1581	1581	1581
AIC	17734.15	17327.1	20762.93
BIC	17744.88	17337.83	20773.66

95% confidence intervals in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.